# DELIVERY DELAY
# & CUSTOMER LOYALTY

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KEY

# STEP

olist

1 PURPOSE & BUSINESS PROBLEMS

2 ABOUT DATA SOURCE

3 DATA STORYTELLING

4 CONCLUSION

5 RECOMENDATIONS

# WHAT ARE WE TALKING ABOUT?

# PURPOSE & BUSSINES PROBLEMS

1. How big is the delivery delay problem?
2. How do customers react to late delivery?
3. Does late delivery affect repeat purchase?
4. Does late first delivery reduce long-term loyalty?

This project analyzes how delivery delay affects customer reviews and repeat purchases in Olist e-commerce.

The focus is on identifying logistics bottlenecks and understanding the impact of the first delivery experience on customer loyalty.

# ABOUT
# DATA SOURCE

This project uses the Brazilian E-Commerce Public Dataset by Olist.
The full dataset contains 9 tables from 2016–2018.

To keep the analysis focused, this study uses only data from 2018 and 6 core tables:

- Orders – order time and delivery status
- Customers – customer location and unique ID
- Order Items – product price and shipping cost
- Products – product category
- Reviews – review score and customer comments
- Payments – payment value, type, and installments

# DATA STORYTELLING

PERCENTAGE

# ON-TIME VS LATE ORDERS

92.3% of orders were delivered on time or earlier than the estimated date, while 7.7% were delivered late. Overall, Olist's logistics performance looks strong based on SLA compliance.

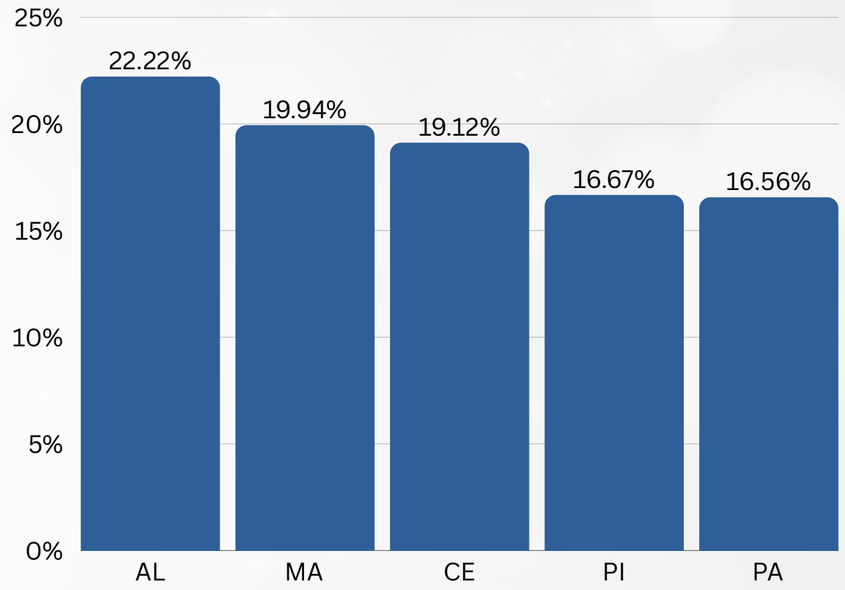
However, even though late deliveries are a small share, their business impact is significant and should not be ignored.

# LOGISTICS

# SLA DECOMPOSITION

| Variabel | Mean | Median |
|---|---|---|
| Approve Days | 1 | 1 |
| Handling Days | 3 | 2 |
| Transit Days | 10 | 7 |
| Delivery Days | 13 | 10 |

Payment approval is fast (1 day) and not a problem. Seller handling is varied (3 days) and can cause small delays. The longest and most unstable stage is transit (9 days). Courier delivery is the main bottleneck.



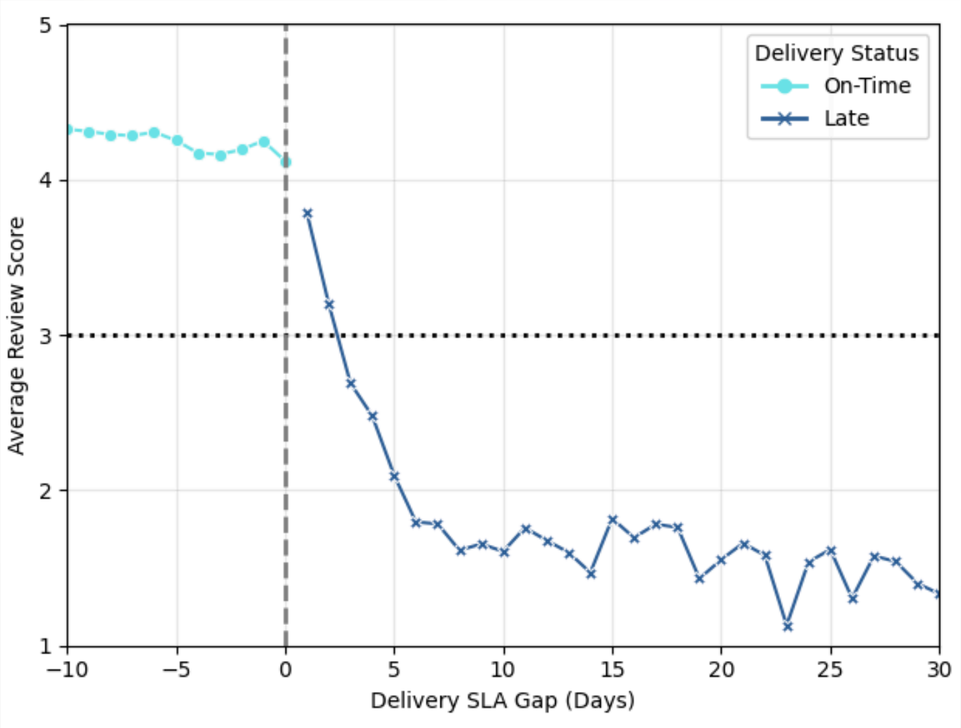

The highest late delivery rates are in AL, MA, CE, PI, and PA. These states have long delivery time (around 20–26 days) and late delivery above 16%. This shows that distance and geography affect delivery more than order volume. States far from main hubs have higher delay risk.

# TOP 5 STATES

# BY LATE DELIVERY

# CUSTOMER **SATISFACTION**

This chart shows the repeat purchase rate based on the first delivery experience. On-time delivery has a 2.1% repeat rate, while late delivery has 2.3%. This does not mean late delivery increases loyalty. The difference is influenced by order volume and sample size. Overall, repeat purchase is very low (<3%), showing Olist is a transactional marketplace, not loyalty driven.



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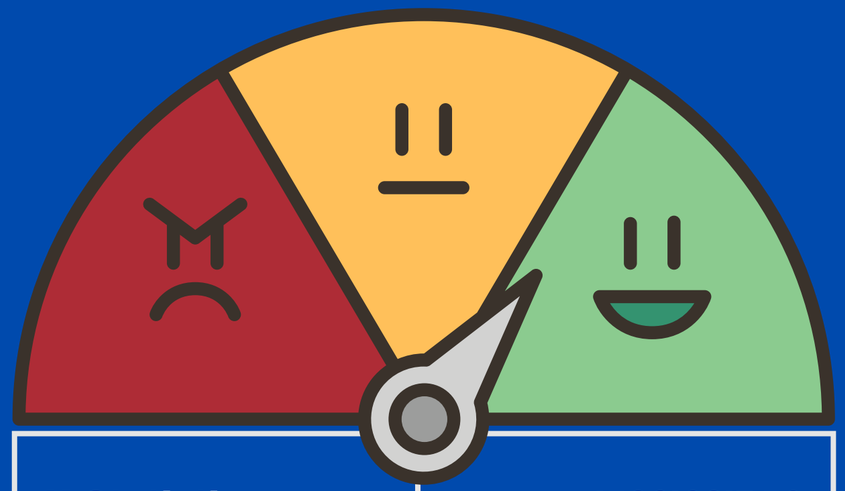


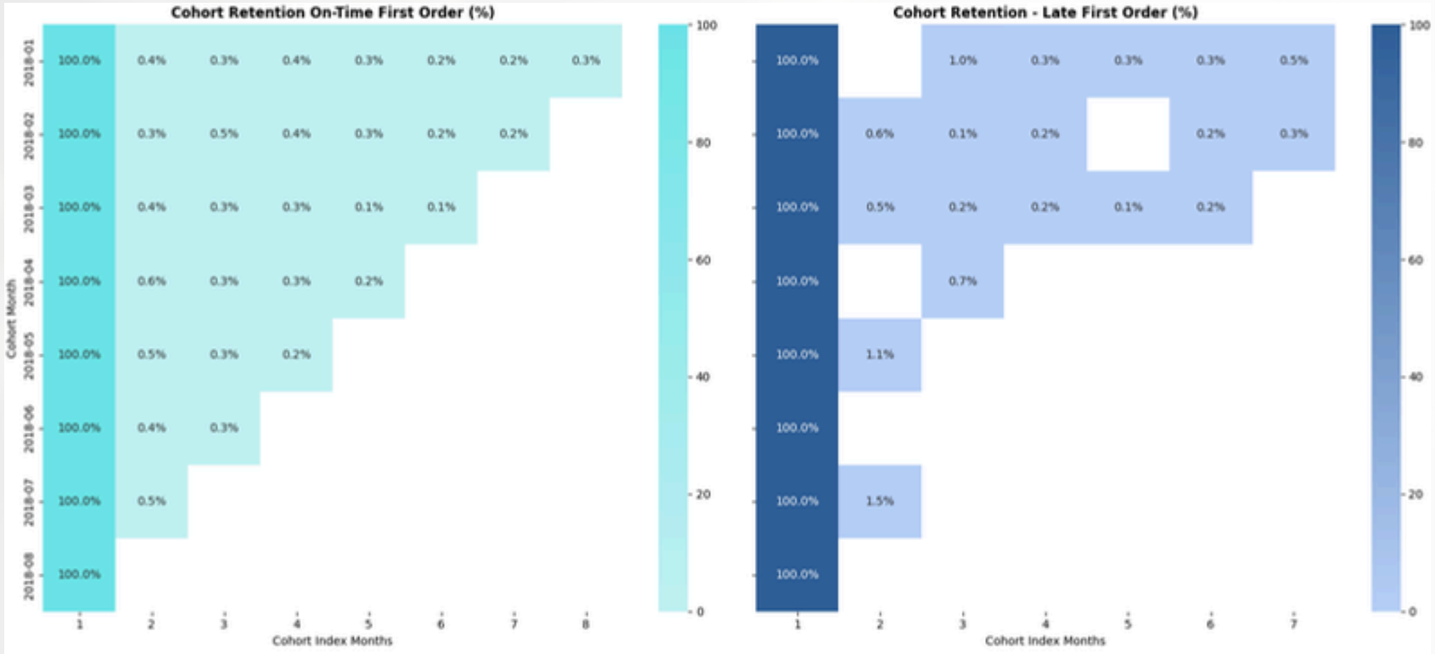
| | |
|---|---|
| **U-Statistic** | 159616917.50 |
| **P-Values** | 0.00 |
| **Median On-Time** | 5.0 |
| **Median Late** | 1.0 |

# MANN WHITNEY U **TEST**

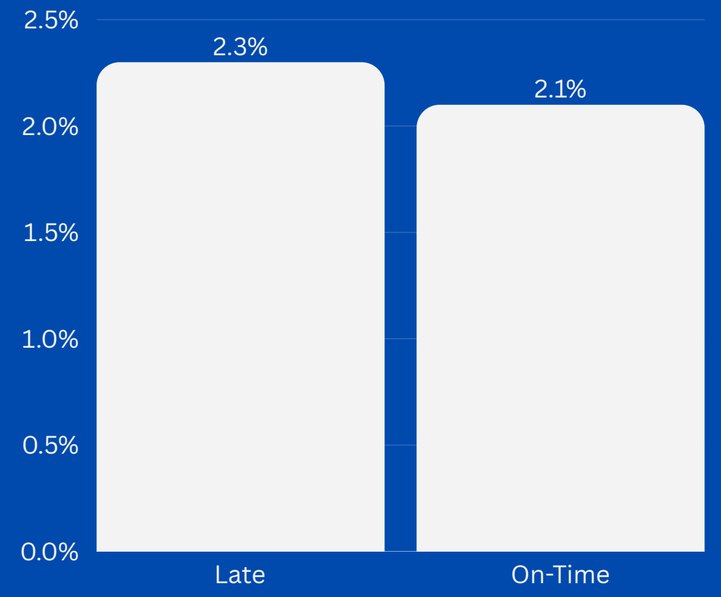
The Mann Whitney U Test compares review scores between on-time and late deliveries. Results show a median score of 5 for on-time and 1 for late delivery, with p-value < 0.05.

This means the difference is statistically significant. Late delivery strongly lowers customer satisfaction.

# COHORT ANALYSIS
# REPEAT PURCHASE




Overall retention is very low (below 1% per month), showing that Olist is mostly a transactional business. However, customers with an on-time first order show a more stable and consistent retention pattern. Customers with a late first order drop faster after the first month. This shows that the first delivery experience matters and late delivery increases churn risk.




This chart shows the repeat purchase rate based on the first delivery experience. On-time delivery has a 2.1% repeat rate, while late delivery has 2.3%. This does not mean late delivery increases loyalty. the difference is influenced by order volume and sample size. Overall, repeat purchase is very low (<3%), showing Olist is a transactional marketplace, not loyalty-driven.

## REPEAT PURCHASE RATE

# CONCLUSION

The analysis shows that even though only 7.7% of orders are late, the business impact is very strong. Most delays happen in the delivery (courier) stage and cause a big drop in review scores.

Statistical, sentiment, and cohort analysis all show that bad delivery experience, especially on the first order, increases churn risk and reduces customer loyalty.

Repeat purchase rate is low in general, but customers with on-time delivery have more stable retention. This proves that logistics performance is very important for customer satisfaction and Customer Lifetime Value (CLV).

STRATEGIC

# RECOMENDATIONS

Even though only 7.7% of orders are late, the impact on customer satisfaction is very strong. The median review score drops from 5 to 1, and the difference is statistically significant.

Olist should focus on managing customer emotions, especially during the first order. This can be done through early delay notifications, clear and empathetic messages, and small compensation before customers feel disappointed.

By managing expectations and emotions, Olist can reduce negative reviews and keep customer trust even when delivery is late.

# THANK YOU

VISIT ME

Happy-Code-byFarhah

Farhah S.E.I.